A GEOMETRICALLY IRREDUCIBLE OPEN OF A NORMAL FIBRE THICKENS TO GEOMETRICALLY INTEGRAL FIBRES AFTER AN ELEMENTARY ETALE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13326-13343 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13345-13362, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Workhorse of the chapter's étale-local-geometry-of-fibres series; the proof's three named inputs are each used exactly once and the final irreducibility step is a cited structural lemma.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas fibre-geometrically-connected-reduced (the connected-neighbourhood engine) and normal (stability of the normal property under base change); Properties, lemma normal-Noetherian (a normal Noetherian scheme with connected underlying space is irreducible).

Source reference. Original lemma 055W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13193–13202. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0391: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5538–5545. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 033M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13362–13362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $h : Y \to S$ be a morphism of schemes. Let $s \in S$ be a point. Let $T \subset Y_s$ be an open subscheme. Assume*

(1) *h is of finite presentation,*
(2) *h is normal, and*
(3) *T is geometrically irreducible over $\kappa(s)$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Then we can find an affine elementary étale neighbourhood $(S', s') \to (S, s)$ and a quasi-compact open $V \subset Y_{S'}$ such that

(a) *all fibres of $V \to S'$ are geometrically integral,*
(b) *$V_{s'} = T \times_s s'$.*

Proof. Apply Lemma 055W to find an affine elementary étale neighbourhood $(S', s') \to (S, s)$ and a quasi-compact open $V \subset Y_{S'}$ such that all fibres of $V \to S'$ are geometrically connected and $V_{s'} = T \times_s s'$. As V is an open of the base change of h all fibres of $V \to S'$ are geometrically normal, see Lemma 0391. In particular, they are geometrically reduced. To finish the proof we have to show they are geometrically irreducible. But, if $t \in S'$ then V_t is of finite type over $\kappa(t)$ and hence $V_t \times_{\kappa(t)} \overline{\kappa(t)}$ is of finite type over $\overline{\kappa(t)}$ hence Noetherian. By choice of $S' \to S$ the scheme $V_t \times_{\kappa(t)} \overline{\kappa(t)}$ is connected. Hence $V_t \times_{\kappa(t)} \overline{\kappa(t)}$ is irreducible by Properties, Lemma 033M and we win. □